



Tracking and Counting Method for Tomato Fruits Scouting Robot in Greenhouse

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Abstract. The quantity of tomatoes is closely related to their yield information, and a powerful inspection robot that can automatically count tomatoes is urgently necessary for the hot and harsh environment of greenhouses. With the continuous progress of computer vision technology, the use of deep learning algorithms for counting tomatoes can greatly improve the inspection speed of the inspection robot. This paper propose a tomato fruit counting method for greenhouse inspection robots, which tracks the position of tomatoes in the image by the spatial displacement information of the robot, while 3D depth filtering can effectively avoid the interference of background tomatoes on the counting. The main advantages of this method are: (1) it can realize the tracking of bunched fruits and the counting of single fruits at the same time; (2) it can avoid the interference of background tomatoes. The experimental results of the greenhouse showed that the accuracy rates of bunch and single fruit counting were higher than 84% and 86% respectively, which greatly improved the inspection speed compared with manual counting and basically meet the counting requirements of the current greenhouse.

Keywords: Greenhouse scouting robot · Tomato tracking and counting · YOLOV5

1 Introduction

Tomato is one of the three major world trade vegetables and occupies an important position in the global vegetable trade. China is the largest tomato producing country in the world, but the production per unit area in China is very low, which is in contrast to developed countries. In recent years, in order to increase the yield per unit area of tomato, it is very important to manage tomatoes scientifically, in addition to using greenhouse cultivation. The use of inspection robots to obtain tomato growth data instead of manual labor can greatly reduce labor costs and improve tomato growing efficiency. The number of tomatoes as an important characteristic of tomato growth process, counting greenhouse tomatoes can not only reasonably arrange the picking cycle, but also make marketing plans in advance to improve economic efficiency.

With the continuous development of computer vision technology and deep learning algorithms, robots are gradually used in agricultural scenes to improve crop growing

efficiency and reduce manual growing costs [1, 2]. The use of computer vision technology instead of the human eye for fruit counting is very widely used in scenes such as orchards and greenhouses.

Suchet Bargoti [3] et al. at the Field Robotics Centre, University of Sydney, Australia presented an image processing framework for fruit detection and counting using orchard image data, including two feature learning algorithms: multiscale multilayered perceptrons (MLP) and convolutional neural networks (CNN), finally achieved the best detection F1-score of 0.861 with WS detection. Xu Liu [4] et al. from the GRASP lab at the University of Pennsylvania presented a novel fruit counting pipeline that combines deep segmentation, frame to frame tracking, and 3D localization to accurately count visible fruits across a sequence of images, the model had an L1 loss of 203, finally achieved an error mean of -0.2% and a standard deviation of 7.8% after the 3D reconstruction correction step. P.J. Ramos [5] et al. from the National Coffee Research Center of Colombia proposed a hand-held non-destructive coffee bean counting method, which can achieve coffee bean counting in some areas through image segmentation algorithms and edge processing methods, which linked the fruits that were counted automatically to the ones actually observed with an R^2 higher than 0.93 one-to-one. Nicolai Häni [6] et al. from University of Minnesota proposed a new fruit detection and counting method based on semantic segmentation and a modular end-to-end system for apple orchard yield estimation, achieved yield estimation accuracies ranging from 95.56% to 97.83%. Many Afonso [7] et al. from Wageningen University used the MASK-RCNN algorithm to accurately identify tomato fruits and classify their maturity, and found that MASK-RCNN can learn the depth data of objects, finally achieved 91% detection accuracy in a single image. Raymond Kirk [8] et al. from University of Lincoln presented a novel multi-object, multi-class fruit tracking system to count fruit from image sequences.

In this study, we propose a tomato fruit counting method for a greenhouse inspection robot, which tracks the position of tomato in the image through the spatial displacement information of the inspection robot. The method is mainly divided into three steps: (1) use the trained fruit detection model to identify the bunch of fruits and single fruits in the picture, and each bunch of fruits will generate an information list; (2) process the tomato information list and delete duplicate labels; (3) convert the list of information to the number of tomatoes.

2 Material and Method

2.1 Tomato Sample and Scouting Robot

In this study, the tomato we selected is a cherry tomato named “jiaxina” from nijiawan greenhouse, Xiangcheng District, Suzhou City, Jiangsu Province. As shown in Fig. 1(a), the tomatoes are planted in a ridge and its fruit grow in a bunch. After artificial agronomic treatment, the mature tomato fruits were in a fixed height range from the ground. There is a track between the two ridges of tomatoes, which is used for vehicles such as greenhouse scouting robot and tomato picking robot. As shown in Fig. 1(b), the scouting robot which we currently use is mainly composed of a depth camera (Realsense D435i), a lifting pole and a greenhouse track chassis. The lifting rod can meet the inspection requirements of tomato fruits with different heights. The chassis contains a set of RFID system. When

the tags at both ends of the track were read, it can control the opening and closing of the inspection function. At the same time, the chassis can move on the track at a constant speed.

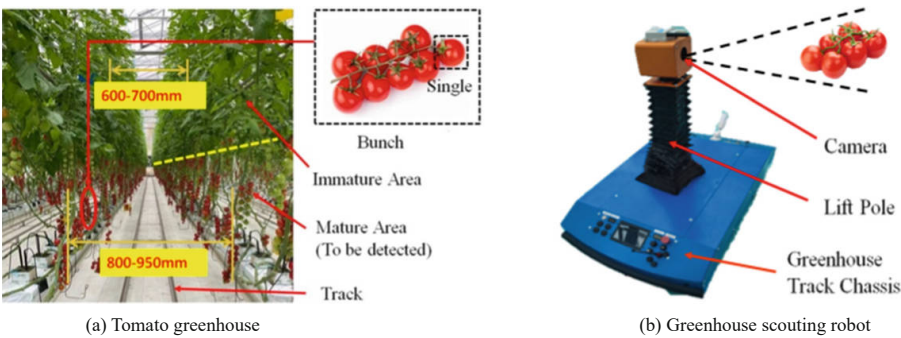


Fig. 1. Tomato greenhouse environment and scouting robot.

2.2 Image Acquisition

The images of tomatoes were acquired by a depth camera mounted on the scouting robot, and the images consist of RGB images of tomatoes and depth images aligned with them, and both the color and depth images were 1280 * 720 pixels. Special attention should be paid to the fact that the position of the camera relative to the scouting robot was fixed during the scouting robot movement. As shown in Fig. 2, the camera axis was perpendicular to the monopoly plane, the camera was 500–650 mm away from the ridge plane, and the height of the camera was 1650 mm above the ground, so that the tomatoes in the mature were appear exactly within the field of view of the camera.

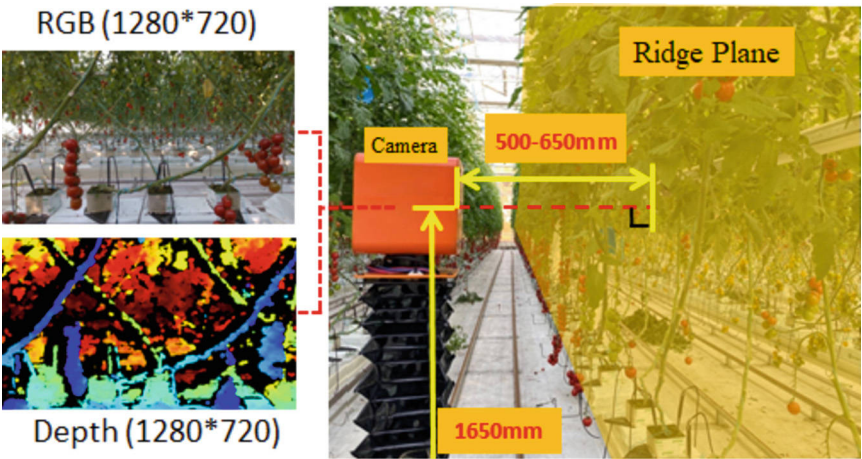


Fig. 2. Position of depth camera.

2.3 Tomato Fruit Detection

Yolov5 is a single-stage target detection algorithm, which has been updated in many iterations and has great superiority in speed and accuracy. In this study we choose YOLOV5 [9] as the tomato fruit detection algorithm. A sufficient amount of dataset should be trained before tomato detection, and we used the scouting robot to drive on the track at a uniform speed (0.1 m/s) and saved color pictures at 5 frames per second. To meet the diversity of samples, we selected data samples from October 2021 to December 2021 for three time periods: 6:00–8:00 AM, 12:00–14:00 PM, and 16:00–17:00 PM, which include different weather lighting environments of sunny, cloudy, and rainy days. After nearly three months of data collection, a total of about 15, 000 images were obtained, and finally 3000 images containing different backgrounds and different fruit ripeness were selected.

In this study we need to train two models, one for the detection of bunches of fruits and one for the detection of single fruits. The filtered images were labeled twice using Labelme software, once with the bunch fruit in the image and once with the single fruit in the image. The labeled images were trained using yolov5. In this study, the ratio of training set, verification set and test set is 1:1:19.

Once the model was trained, we can detect the tomatoes in the greenhouse. As shown in Fig. 3, the detection of tomatoes was divided into 3 main steps: the first step detects tomato bunches, the second step detects the number of individual fruit in each bunch, and the third step assigns an information list to each tomato bunch. Finally, we used the list $[frame, ID, x, y, w, h, a, D, s]$ to mark each tomato bunch detected, $frame$ represents the number of frames of the current image, the ID represents the total number of tomato bunches detected from the beginning to the present, x, y, w, h represent the pixel coordinates of the center point and the length and width of the target rectangle box obtained from the tomato bunch detected, a represents the number of single fruit of the current tomato bunch after single detection, D represents the average value of the depth of each single fruit in the current bunch, and s represents the accumulated distance traveled by the scouting robot on the track from the beginning of detection to the present.

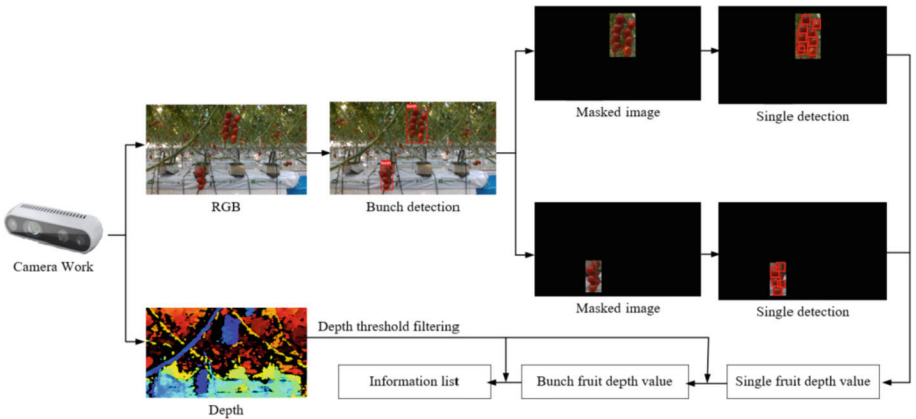


Fig. 3. Flow chart of tomato fruit detection.

The main use of the depth map in the fruit detection process was to filter the background mis-detected tomatoes after getting the depth values of bunch and single fruit. In order to prevent the following tracking error caused by the incomplete fruit of the tomato bunches at the edge of the image, we set invalid areas on both sides of the image, if x is less than 100 or greater than 1180, then the list of information obtained will be deleted.

2.4 Tracking and Counting of Tomato Fruit

After the fruit detection step, the tomatoes detected in each frame of the picture will correspond to an information list. During the inspection process, the same bunch of tomatoes may be identified in multiple frames of the image, which means that a bunch of tomatoes may correspond to multiple lists. Tracking counting is the process of removing these duplicate labels. As is shown in Fig. 4, take frame n and frame $n + 1$ as examples, at this time, the scouting robot walks s_n and s_k respectively. It is assumed that three strings of tomatoes are found in each frame.

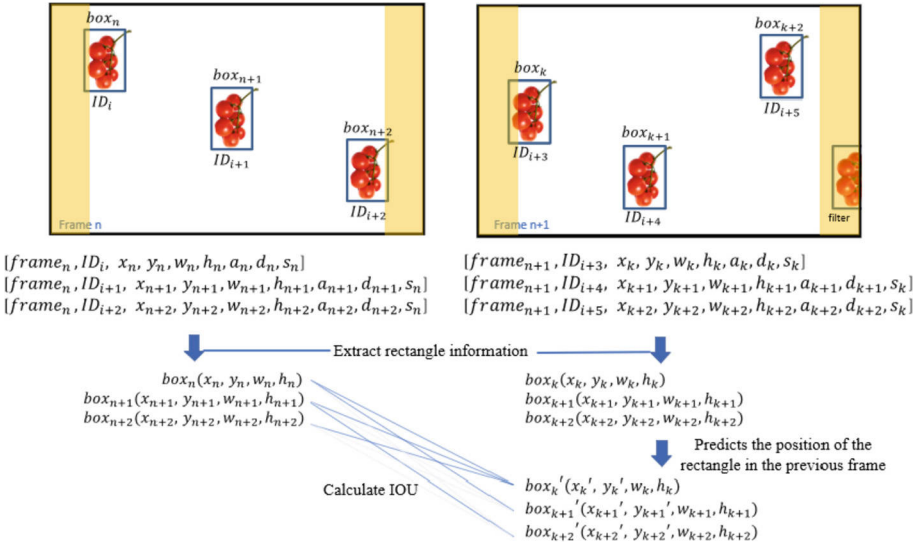


Fig. 4. Schematic diagram of tomato tracking.

In this study, we used the information of tomato detection rectangle in the current frame and the shift information of the robot to predict the position of the rectangle in the previous frame image, and then the predicted rectangle box and all the rectangle boxes obtained from the previous frame were calculated as the intersection union ratio (IOU) [10]. If IOU calculated was within the set threshold, then the information list corresponding to this rectangle box should be deleted. As shown in Fig. 4, ID_{i+3} tomatoes in frame $n + 1$ should be taken as an example to introduce the prediction process of the rectangular box.

The rectangular box corresponding to the information list of ID_{i+3} tomato is box_k . First, converted the pixel coordinates (x_k, y_k) of the center point of the box to the coordinates (X_k, Y_k) under the camera coordinate system. In Eq. (1) and Eq. (2), c_x, c_y, f_x, f_y are camera internal parameters.

$$X_k = \frac{d_k * x_k - c_x * d_k}{f_x} \quad (1)$$

$$Y_k = \frac{d_k * y_k - c_y * d_k}{f_y} \quad (2)$$

The moving distance of the scouting robot from frame n to frame $n + 1$ is calculated by the cumulative displacement difference (Eq. (3)). In Eq. (4) and Eq. (5), x'_k and y'_k respectively represent the pixel coordinates of the center point predicted in frame n of box_k , finally, x'_k and y'_k are calculated (Eq. (6) and Eq. (7)).

$$s = s_k - s_n \quad (3)$$

$$X_k - s = \frac{d_k * x'_k - c_x * d_k}{f_x} \quad (4)$$

$$Y_k = \frac{d_k * y'_k - c_y * d_k}{f_y} \quad (5)$$

$$x'_k = \frac{d_k * s_k - s * f_x}{d_k} \quad (6)$$

$$y'_k = y_k \quad (7)$$

After predicting the position of box_k in frame n , we calculated the IOU of the predicted new box'_k and all rectangular boxes of frame n in turn. If the IOU calculated was greater than 0.8, the information list corresponding to box_k was considered to be a duplicate list. On the contrary, if the IOU calculated was less than 0.8, the list should be retained. In order to prevent the tomato in a frame image from being successfully detected, each list must be calculated with all the information lists in the previous 30 frames until it is considered as a duplicate list.

2.5 Result Output

After patrol inspection, the number of information lists retained is the total number of tomato bunches, and the sum of value a in each information list is the total number of single tomato fruit. At the same time, the images of each tomato bunch and the corresponding information list were saved as patrol inspection record.

3 Result and Discussion

As shown in Table 1, in order to verify the reliability of the algorithm, we selected three tracks in different areas of the greenhouse to count the tomato fruits, and compared the number of fruits obtained with the actual number of fruits obtained by manual counting. It was found that the algorithm has high accuracy in tomato fruit counting. From the experimental results, the accuracy of single fruit counting is lower than that of bunch fruit counting, which is due to the problem of mutual shading of single fruits. Although the inspection process can identify single fruits from different angles, the shading between fruits will still affect the counting.

Table 1. Experimental results of counting tomatoes by described method.

	test1	test2	test3
Speed	0.1 m/s	0.1 m/s	0.1 m/s
Actual number of bunch fruits	30	68	86
Number of detected bunch fruits	34	75	98
Accuracy of bunch fruit counting	86.7%	91.2%	87.2%
Actual number of single fruit	356	664	704
Number of single fruit detected	302	586	593
Accuracy of single fruit counting	84.8%	88.3%	84.2%

In addition, we set up a group of comparative experiments, using the current mainstream crowd counting algorithm (Deepsort) to count the tomatoes in the greenhouse. Deepsort [11] is a classical multi-target tracking algorithm based on Tracking by Detection strategy. Its main principles are Hungarian algorithm and Kalman filter. We used the color video stream recorded by the inspection robot on the same track at the same time to do experiments. First, the trained yolov5 model was used to identify the bunch fruit in each frame of images, then deepsort tracked the position of the bunch fruits in the image, and the final output detection results are shown in Table 2. In the tracking process, each tracked bunch fruit was given a label. We cut out the bunch fruit rectangle of the same label for single fruit detection, so that we can get the number of single fruit of each bunch fruit.

From Table 2, we can find that the accuracy of bunch fruits counting of deepsort is lower than that of Table 1, but the number of detected tomatoes is higher than that of Table 1. The reason is that deepsort does not have the function of deep filtering, and many background tomatoes are included in the final number. Comparing the two counting methods, it is easy to find that the number of bunches of fruit counted is higher than the actual number. Because the bunch tomatoes were not successfully identified in several consecutive frames during the process of being tracked, and when they were identified again they were no longer successfully tracked, and then the algorithm defined them as newly identified bunches of fruit, so that the number of bunches of fruit is higher than

the actual number. Therefore, improving the accuracy of target recognition in a single frame image is very important for target tracking and counting.

Table 2. Experimental results of counting tomatoes by Deepsort algorithm.

	test1	test2	test3
Speed	0.1 m/s	0.1 m/s	0.1 m/s
Number of detected bunch fruits	49	116	144
Accuracy of bunch fruit counting	61.2%	58.6%	59.7%
Number of single fruit detected	265	586	593
Accuracy of single fruit counting	74.3%	77.4%	79.2%

4 Conclusion and Future Work

We proposed a tomato fruit counting method for a greenhouse inspection robot that uses the spatial displacement information of the inspection robot to track the position of tomatoes in an image to achieve the function of tomato fruit tracking and counting. This method uses a trained fruit detection model to identify each bunch of fruits in the image and calculate the number of single fruits in each bunch. At the same time, 3D deep filtering can effectively avoid the interference of background tomatoes. Then, the displacement information and coordinate transformation of the inspection robot are used to eliminate the repeated list representing the same string of tomatoes. Finally, the retained list can be converted into the quantity information of tomato fruits. Then, the duplicate list representing the same bunch of tomatoes is removed by coordinate transformation and the displacement information of the inspection robot, and the retained list is converted into the quantity information of tomato fruits.

The tomato fruit counting method proposed in this paper has the following two advantages: (1) it can realize the tracking of bunched fruits and the counting of single fruits at the same time; (2) Depth information filtering can effectively reduce the interference of background tomato fruit.

Our future work will mainly focus on how to improve the accuracy of string fruit and single fruit recognition, especially how to solve the occlusion problem of single fruit. We may train and use different models for different periods, or modify the network to improve the recognition accuracy.

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